

MEDICAL MONITORING AND ADVANCED SENSOR DATA PROCESSING – TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

1 General Information

The practical experiments should be done in groups of up to three people. Each group must submit one solution via Digicampus. Of course, you can discuss the tasks between groups, but forwarding/copying other students' source code is forbidden.

Prepare the task until the deadline provided below/in Digicampus.

On Digicampus, there is a template for you to prepare your code for submission. Please complete your code within this template. The **README.md** file provides information on how to set up your Python environment for running your code. A Jupyter notebook example is also provided. You can adapt the content of this Jupyter notebook for preparing your solution.

Before uploading your solution, please make sure that the following requirements are met:

- Prepare your result documentation as Jupyter Notebook and a report
- Print a PDF version of your Jupyter notebook AND add it to the .zip archive you upload on Digicampus.
- All source code must be available and executable for us (this implies that you specify the required packages/libraries. In addition, you have to provide necessary data (ideally data should be in a subfolder, and you use relative paths) and/or give advice if paths have to be adjusted; in order to do so, you should provide a reasonable README file)
- **GREEN** sections in the chapter are tasks. Please include a numbering according to the questions' number Q<xy>
- List your sources (if applicable)

Submission deadline is **Thursday, 03.07.2025, 08:00**; submission is done via **Digicampus**.

Please ask in case of any doubts regarding the contents or the formal procedure.

2 Elements to be delivered

1. Jupyter Notebook: the students must deliver the code (as .ipynb file) they developed to visualize, analyse and plot the data according to the template provided.
A pdf version of the Jupyter notebook with the results of the notebook's run should be provided as a backup in case the notebook fails to be run again.

NOTE:

A requirements.txt file is provided on digicampus. Students HAVE TO USE ONLY the libraries that are provided within this file.

2. The students' recordings that are necessary to execute their Jupyter notebooks.
3. Report: the students must deliver a report (as .pdf file) using the ReportTemplate.docx found in Digicampus. The report must content the following:
Title Page
Title: Clear and specific
Course name / lab name
Students names
Date
Answer to the questions (with the appropriate question numbering)
4. When a question asks to discuss a topic, the word limit is an upper limit.

3 Remarks on the Hardware and software

A documentation to perform the recordings based on the smartwatch and Biopac platform is provided [1]. Executables to perform recordings via the ADI Study watch are provided in a Flashdrive [2]. Those executables should be copied to the computer which will record the data.

3.1 Biopac

One can export the acquired data in a format that is suitable for postprocessing.

3.2 ADI Study smartwatch

The communication between the smartwatch and the computer is established via a Dongle connected to a plugged USB port. One needs to check the port address (COM) using the device manager.

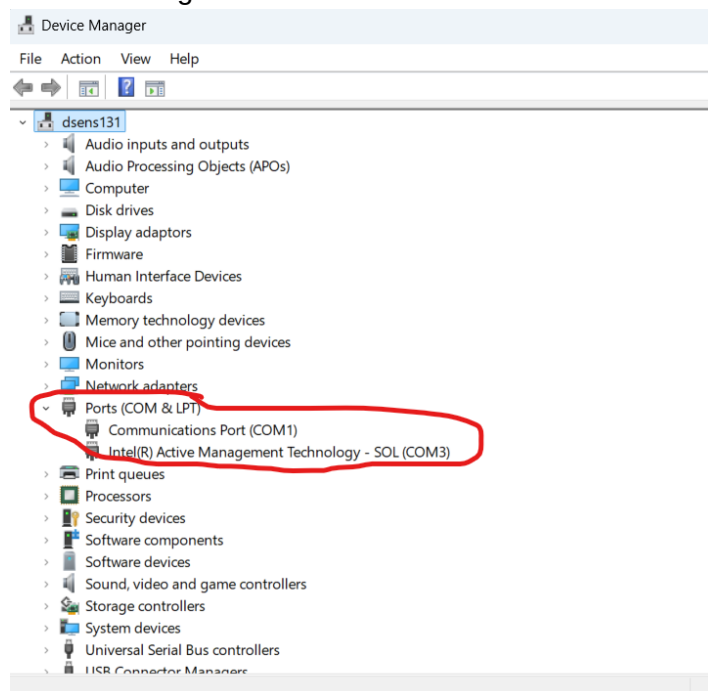


Figure 1 Device manager (on Windows).

The recordings are stored on the computer in .csv files.

The smartwatch can provide an electrocardiographic signal (ECG), a photoplethysmographic signal (PPG), and accelerometer signals (along the 3 directions).

4 Remarks on reports and data collection

Following some remarks on the report and data collection:

- Reports for data gathering should be detailed and complete. Students will use the data they collect to work on it by themselves.
- The report must assure the reproducibility of the experiment and data gathering, bearing in mind the amount and quality of data that is intended to be collected.
- If you cite from the literature, use the APA citation style
- For plots in the time domain, signals must be plotted with x-axis in time, in seconds.
- All plots must have axes labelled in a meaningful way.
- All plots or images must have caption written in a meaningful way.
- In the caption, do not forget to refer to the question number **Q...** the plot is built for.
- If a question asks you to make a plot, include the plots with appropriate reference to the question in the caption, and write a reference to the plot in the report.

For example :

Qxx. Area under the curve obtained on the test set by several classifiers is displayed on Figure 2.

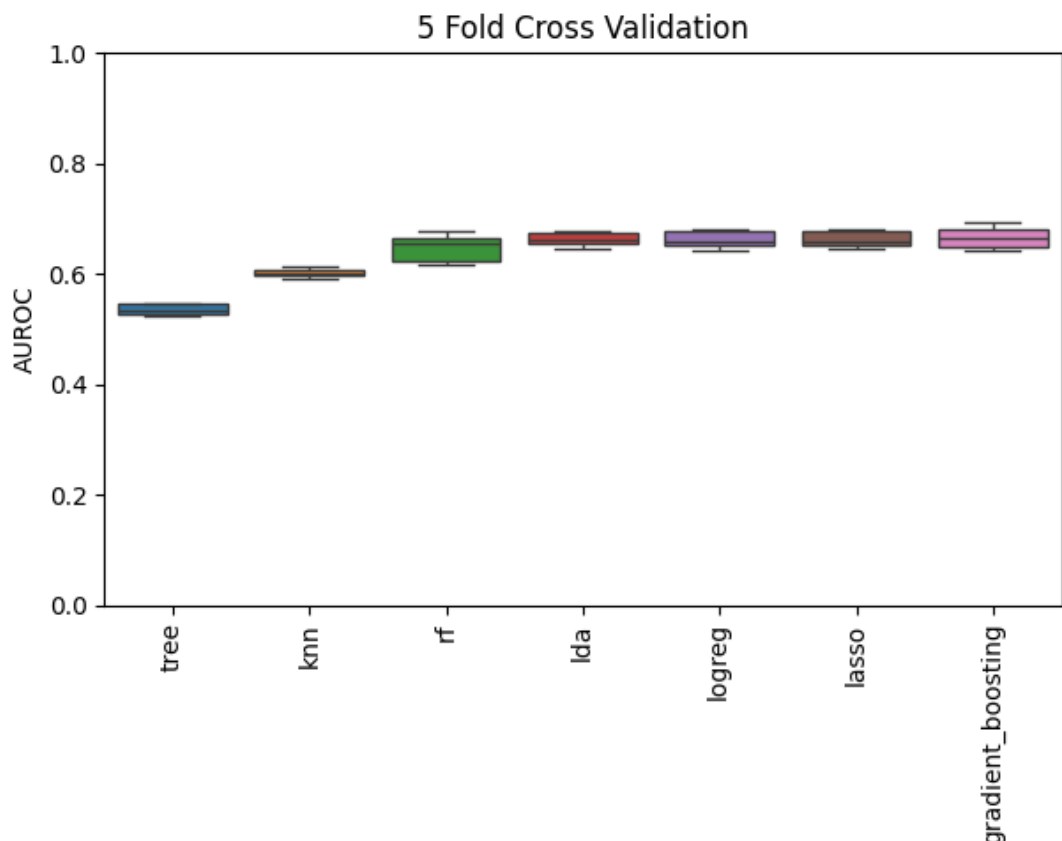


Figure 2 Area under the curve of several classifiers on the test set (**Qxx**).

- **ATTENTION:** Have a look at the general document on our expectation and common mistakes, which is available in Digicampus

5 Tasks

5.1 Recordings

This is the last task to perform own recordings. These recordings will be used for all remaining tasks this semester. Take care that you are doing them thoroughly. The recordings will be done under varying condition, rest, movement and exercise. Watch out that sensors and cables are attached to your body in order that they won't be affected too much by your movements (for example fix the biopac ECG measurement device through your belt loops and attach the cable of the Biopac PPG Sensor with adhesive strips to your skin). It would be best if all recordings are done by one person of your group. The person doing the exercise should try to avoid talking/laughing and solely do the exercises. The other persons should help in order to inform about measurement times and when to switch the exercise. They should mark when a specific exercise is started/ended with biopac markers (e.g. press F7). That way you can distinguish better between different exercises later. Also, while walking the stairs, they have to make sure the person is not going too far away from the measurement laptop and biopac. You can not walk away more than one floor from the measurement system. You might do a test measurement and see how far the connection stays stable for you.

For this exercise you need to perform the following recordings. For all recordings you use Biopac (PPG and ECG) and the smartwatch (PPG and Acceleration data) simultaneously. PPG signals (wrist and finger) are to be recorded at the same hand. Make sure that you produce an artifact at the same time in both biopac and smartwatch, which will be used to synchronize the signals later and mark the start of your measurement task. Between "Stairs", "Workout" and "Resting" you can do small breaks. Those are independent measurements.

- Stairs
 - Rest (stand still or sit) for 5 minutes
 - Walk the stairs up and down for 10 minutes straight, no rest. (do it slowly enough in order to be able to complete 10 minutes of stair climbing)
 - Rest (stand still or sit) for 5 minutes
 - Total of 20 min measurement
- Workout (if you can't do the workout, you can try to do easier exercises (**talk to us and document the changes in your report!**))
 - Do the following workout for 2 rounds straight
 - 30s squats, 60 s rest (stand still)
 - 30s walking stairs up and down, 60 s rest (stand still)
 - 30s jumping jacks, 60 s rest (stand still)
 - 30s rotate your wrist, 60 s rest (stand still)
 - 30s stretch out both arms to the side and do small circles, 60 s rest (stand still)
 - Total of 15 min measurement
- Resting
 - Measure 30 min data only during rest
 - This data will only be used in task 5 but has to be recorded now
 - You don't have to record 30 min straight. You can do for example 3 x 10 min or 6 x 5 min
 - You should try to avoid talking/laughing during measurements

5.2 Questions

Q1 (0.5 Points) Consider the data “Stairs”. Resample the smartwatch signals to 1000 Hz and synchronize all signals using the artifact you created at the beginning of your measurement. Cut the signals in order that it only contains the exercise. Plot the PPG (biopac), PPG (smartwatch), ECG (biopac), Acceleration data (smartwatch) each in an own plot (4 plots) with corresponding time axis. Mark the transition between walking and resting in your plots.
Note: The artifact from the beginning is removed and all signals (from one measurement) have the same length. It might make sense to write a method for this Question

Q2 (0.5 Points) Consider the data “Workout”. Resample the smartwatch signals to 1000 Hz and synchronize all signals using the artifact you created at the beginning of your measurement. Cut the signals in order that it only contains the exercise. Plot the PPG (biopac), PPG (smartwatch), ECG (biopac), Acceleration data (smartwatch) each in an own plot (4 plots) with corresponding time axis. Mark the transitions between exercise and resting.
Note: The artifact from the beginning is removed and all signals (from one measurement) have the same length.

Q3 (0.5 Points) Consider the data “Stairs”. Use the neurokit2 toolbox [3] to perform R-peak detection of the ECG.
Then calculate the instantaneous heart rate (i.e. the heart rate for each detected beat) without (!) the help of the toolbox.
Calculate a smoothed heart rate by taking the *median* per 10s window sliding it by 1s and assign the calculated value to the middle of the window. That way you have one heart rate per second.
Plot a 20s segment of the ECG with the corresponding R-peak detections, where the detection worked well.
Plot a 20s segment of the ECG with the corresponding R-peak detections, where the detection was faulty.
Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.
Discuss differences that occur between the two heart rates (10 sentences max)
Note: it might make sense to write a method for this because later you'll need both heart rates for all measured signals.

Q4 (0.5 Points) Consider the data “Workout”. Use the neurokit2 toolbox [3] to perform R-peak detection of the ECG.
Then calculate the heart rate for each detected beat without(!) the help of the toolbox.
Calculate a smoothed heart rate by taking the *mean* per 10s window sliding it by 1s and assign the calculated value to the middle of the window. That way you have one heart rate per second.
Plot a 20s segment of the ECG with the corresponding R-peak detections, where the detection worked well.
Plot a 20s segment of the ECG with the corresponding R-peak detections where the detection was faulty.
Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.
Do you notice any differences between the two heart rate courses?

Q5 (0.5 Points) Consider the data “Stairs” and use the smartwatch PPG data. Use the neurokit2 toolbox [3] to perform PPG peak detection.

Then calculate the heart rate for each detected beat without(!) the help of the toolbox.

Calculate a smoothed heart rate by taking the *mean* per 10s window sliding it by 1s and assign the calculated value to the middle of the window. That way you have one heart rate per second starting at 5s.

Plot a 20s segment of the PPG with the corresponding beat detections, where the detection worked well.

Plot a 20s segment of the PPG with the corresponding beat detections, where the detection was faulty.

Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.

Do you notice any differences between the two heart rate courses?

Q6 (0.5 Points) Consider the data “Stairs” and use the biopac PPG data. Document which one you are using in order to allocate the data to the correct tracks above. Use the neurokit2 toolbox to perform PPG peak detection.

Then calculate the heart rate for each detected beat without (!) the help of the toolbox.

Calculate in 10s windows sliding it by 1s the *mean* heart rate and assign the calculated value to the middle of the window. That way you have one heart rate per second starting at 5s.

Use the same segments as in Q5:

Plot a 20s segment of the PPG with the corresponding beat detections, where the detection worked well.

Plot a 20s segment of the PPG with the corresponding beat detections, where the detection was faulty.

Plot the course of the instantaneous heart rate over time together with the course of the smoothed heart rate where you used sliding windows.

Do you notice any differences between the courses in Q5 and Q6?

Q7 (1 Point) Consider the smartwatch PPG and choose either “Workout” or “Stairs” data for this Question. Document which one you are using in order to allocate the data to the correct tracks above. Use the same 10s windows and sliding as before, but this time for each window identify the heart rate in the frequency domain, i.e. do a fast fourier transform for heart rate extraction. Before transforming the window into the frequency domain you should do some preprocessing for the 10 s window (e.g. highpass filtering at a meaningful frequency (e.g. using for example butterworth) or detrending). Then find a way to algorithmically detect the heart rate in the spectrum and assign the detected heart rate to the middle of the window. That way you have one heart rate per second starting at 5s. Describe how you did your preprocessing and heartrate extraction.

Q8 (1 Point) Consider the ECG heart rate being ground truth. Calculate the mean absolute error (MAE) and the Pearson correlation coefficient (*only using numpy for both!*) of the smartwatch PPG HR with and without smoothing for all measurements (You have to do that for all measurements, including those that haven’t been addressed in Q1- Q7 so far). Write down the equations for MAE and person correlation coefficient. Also calculate the MAE and Pearson correlation coefficient for each phase of your exercise (‘walking stairs’, ‘resting’,

'squats', 'jumping jacks', 'rotate wrist', 'circle arms'). Report the quantitative results as table. Discuss the differences that might occur (one paragraph, 15 lines max). Repeat Q8 for the biopac PPG and discuss any differences. Add MAE and Pearson correlation coefficient for the HR extracted from the frequency domain in your table. Which results are the best? Discuss differences and possible reasons (one paragraph, 15 lines max).

Q9 (1 Point) Do some research on Bland Altman plots. Consider the ECG heart rate being ground truth. Plot the biopac PPG heart rate with the ground truth as a Bland Altman plot for both smoothed and not smoothed heart rate and also for "Stairs" and "Workout" data (you can choose one measurement, but document which one). Plot the smartwatch PPG heart rate with the ground truth as a Bland Altman plot for both smoothed and not smoothed heart rate and also for "Stairs" and "Workout" data (you can choose one measurement, but document which one). Also do a Bland Altman plot on the frequency based heart rate extraction. Using one of your Bland Altman plots (you choose) explain the principle of this plot. What is seen and how can you interpret it in this context? (one paragraph, 15 lines max). Discuss differences that might occur in the plots (one paragraph, 15 lines max).

6 Material (and recommended readings)

- [1] Data Recordings- Overview.pdf
- [2] Flashdrive with the executables for the smartwatch
- [3] Neurokit - <https://neuropsychology.github.io/NeuroKit/> Accessed 11 June 2025.